Target Image
Reference Image
Reference Branch
Target Branch
Frozen
Trainable
F_r Reference Feature
F_t Target Feature
Frozen Backbone
F_t
F_r
Reference Mask M_r
RTAdapter
$\tilde{F}_r$
Frozen Decoder
Prototype Decoder
4D Correlation Decoder
SAM prompt Decoder
plug-and-play with multiple method families
Target Mask
RTAdapter: Episode-Testing Reference Feature Adaptation
1) Foreground Adaptation
F_r
M_R
F_r^{bg}
F_r^{fg}
Adapter
optimize foreground only
$\tilde{F}_r^{fg}$
$\tilde{F}_r$
2) Optimization Objectives
i) Distribution Loss L_{EMD}
minimize the distribution distance (EMD) between $\tilde{F}_r$ and F_t
$\tilde{F}_r$
Minimize Distance
F_t
Transport Plan Π
Π_{ij}: mass transport from reference i to target j
μ
ν
Marginal Mass
$\sum_i \mu_i = 1$
$\sum_j \nu_j = 1$
ii) Semantic Alignment Loss $L_{semantic}$
Prototype Level
Priormap Level
Reference Space
original reference prototype
Cosine
adapted reference prototype
adapted F_r^{fg} patch
Π_{ij}/μ_i
Transport weighted target feature
reference priormaps
BCE Loss
reference Mask
Target Space
adapted reference prototype
Cosine
foreground mass weighted target prototype
BCE Loss
Transport branch
pre-adapter target priormap
Adapter branch
BCE Loss
foreground transport mass fraction for target patch j: $\sum_{i \text{ in ref\&fg}} \Pi_{ij}/\nu_j$
adapted target priormap